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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2025-2026

IE2104 – DIGITAL ELECTRONICS

April / May 2026 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) A 10-bit register stores a data 1010111101. What is the equivalent decimal number if it represents:

   (i) a 1's complement number?
   (ii) a 2's complement number?

   (8 Marks)

   (b) Simplify the logic function $y(a,b,c,d) = \sum m\,(0,1,2,3,7,8,9,10,11,15)$ to the minimized sum of products and the minimized product of sums. Prove that both the minimized expressions are equivalent to each other using Boolean Algebra.

   (10 Marks)

   (c) Design a CMOS circuit with minimum number of transistors for the logic function $z(a,b,c,d) = (a+b)\cdot(c+d)$. Label clearly the inputs a, b, c and d, and output $z(a,b,c,d)$ in the CMOS circuit.

   (7 Marks)

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2. (a) Figure 1 shows a 2-bit binary to 3-segment Chinese numeric display decoder with its outputs S_a, S_b and S_c connected to a 3-segment LED display. The function table of this decoder is given in Table 1. Design the decoder using ONLY a 2-to-4 decoder and two-input OR gates. The function table of the 2-to-4 decoder is given in Table 2.

(8 Marks)

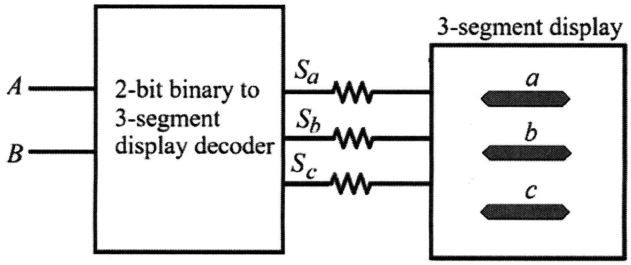


Figure 1

Table 1

Inputs		Outputs		
A	*B*	S_a	S_b	S_c
0	0	0	0	0
0	1	1	0	0
1	0	1	1	0
1	1	1	1	1

Table 2

Inputs		Outputs			
A	*B*	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

Note: Question No. 2 continues on page 3.

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(b) Design a full adder using ONLY a Programable Logic Array (PLA) as shown in Figure 2 and two-input XOR gates. The function table of the full adder is given in Table 3. Draw the final circuit that includes both the PLA and the XOR gates. Label clearly in the circuit the inputs A, B and C_i, and the outputs S_o and C_o.

(9 Marks)

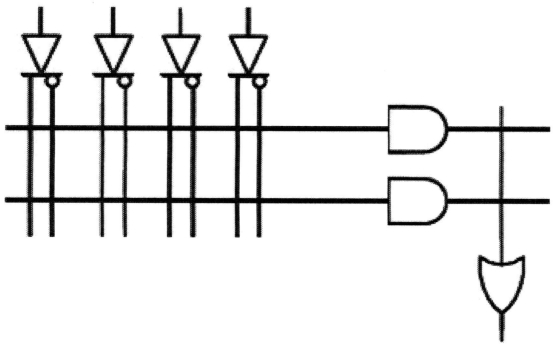


Figure 2

Table 3

Inputs			Outputs	
A	B	C_i	S_o	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

(c) Implement the function $y(a, b, c) = (\bar{a} + b)(a + c)$ using ONLY a 4-to-1 multiplexer (MUX) as shown in Figure 3 on page 4, where its function table is given in Table 4 on page 4.

(8 Marks)

Question No. 2 continues on page 4.

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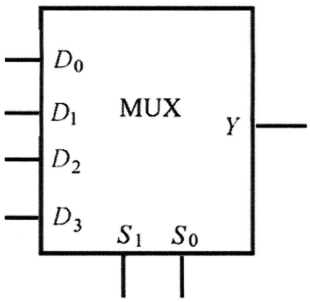


Figure 3

Table 4

Select		Output
S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

3. (a) Figure 4 shows a sequential circuit, where X and OUT are the sequence input and output, respectively, and CLK is the clock signal.

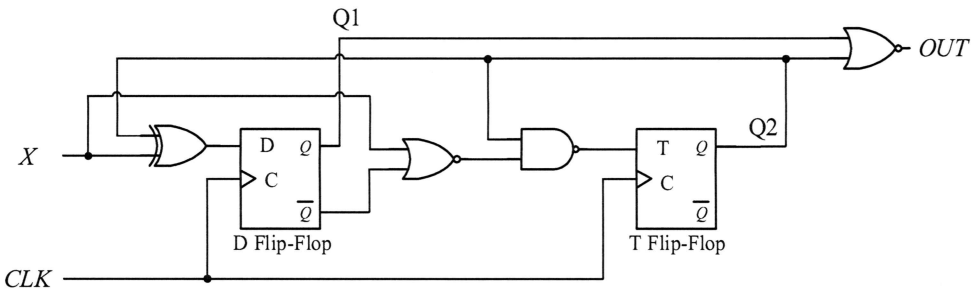


Figure 4

(i) Determine whether the circuit in Figure 4 is a Moore or a Mealy state machine. Briefly explain your answer.

(ii) Write the input functions (excitation functions) of the D and T flip-flops.

(iii) Construct the state transition table.

(iv) Draw the state diagram.

(20 Marks)

Question No. 3 continues on page 5.

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(b) The timing parameters (in ns) of the logic gates and the flip-flops in Figure 4 on page 4 are given in Table 5. Determine the setup and hold time margins if the frequency of *CLK* is 20 MHz. Assume that *X* switches at the rising edge of *CLK*.

(5 Marks)

Table 5

Device	t_d (min)	t_d (max)	t_{su} (setup)	t_h (hold)
XOR	6	12	-	-
NAND	4	6	-	-
NOR	4	6	-	-
T flip-flop	4	10	7	5
D flip-flop	3	6	5	4

4. (a) Draw the state diagram of a 3-bit synchronous counter.

(5 Marks)

(b) Draw the state transition table of the state diagram in Question 4(a) by using:

i) D flip-flops

ii) T flip-flops.

(5 Marks)

(c) Using the state transition tables in Question 4(b), design the 3-bit synchronous counter using D flip-flops. Show detailed steps of derivation and the final circuit diagram.

(10 Marks)

(d) Determine how many address lines and data lines are needed for each of the following memory modules:

(i) 16M × 32

(ii) 8Mbytes

(5 Marks)

END OF PAPER

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IE2104 DIGITAL ELECTRONICS

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.